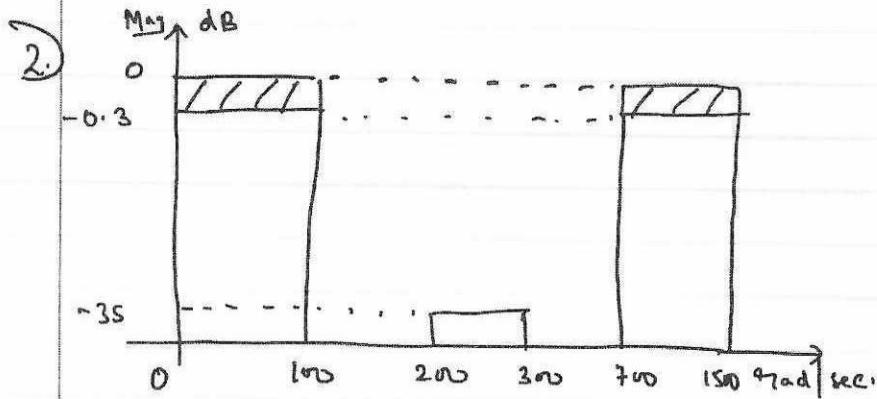




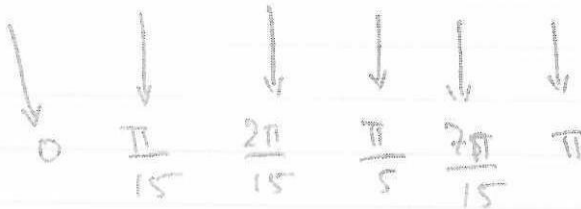
Bandstop filter

$$f_s = \frac{3000}{2\pi} \text{ Hz}$$

$$\Rightarrow T = \frac{\pi}{1500}$$

Soln.

Digital
filter
specs



$$\Omega_c = \frac{2}{T} \tan \frac{\omega_i}{2}$$

ω (radians)

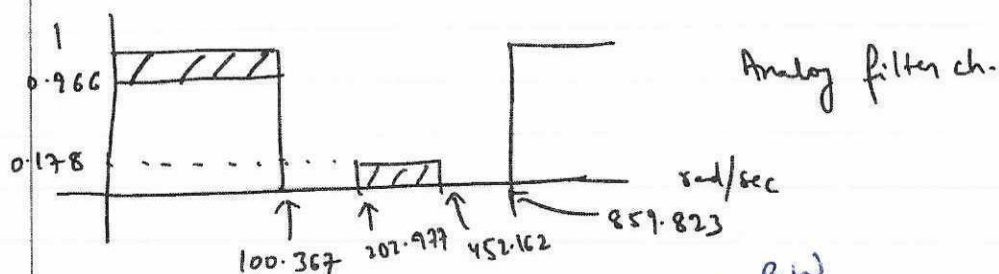
$\Omega \rightarrow$ (rad/sec)

0	$\rightarrow$	0
$\pi/15$	$\rightarrow$	$100.367 (\Omega_{p1})$
$2\pi/15$	$\rightarrow$	$202.977 (\Omega_{s1})$
$\pi/5$	$\rightarrow$	$310.275 (\Omega_{s2})$
$7\pi/15$	$\rightarrow$	$859.823 (\Omega_{p2})$
π	$\rightarrow$	∞

$$\Omega_0^2 = \Omega_{p1} \Omega_{p2} \Rightarrow \Omega_0 = 293.765 \text{ rad/sec}$$

$\sqrt{\Omega_{s1} \Omega_{s2}} = 250.956 \text{ rad/sec}$ $\uparrow$ Geometric symmetry does not exist
 $\therefore$ this is less than $\Omega_0 \therefore \Omega_{s2} \rightarrow$ changed

$$\omega'_{s2} = \frac{(293.765)^2}{\omega_{s1} = 202.977} = 452.162 \text{ rad/sec}$$



CPE
design

Normalised
Stopband
freq

$$\omega_s = \left(\frac{\omega_{p2} - \omega_{p1}}{\omega'_{s2} - \omega_{s1}} \right) = 3.418 \quad \text{BW}$$

$$N_c \geq \frac{\cosh^{-1} \sqrt{\left(\frac{1}{0.0178} \right)^2 - 1}}{\cosh^{-1} \sqrt{\left(\frac{1}{0.966} \right)^2 - 1}} = 3.178$$

$$\cosh^{-1} 3.418$$

$$N_c = 4$$

NOTE

$$H_a(s) = \frac{c_1 c_2 \omega_p^4}{(s^2 + b_1 \omega_p s + c_1 \omega_p^2) (s^2 + b_2 \omega_p s + c_2 \omega_p^2)} \times 0.966$$

$$\omega_p \rightarrow 1 \quad (\text{Normalized filter})$$

$$\Rightarrow \frac{1}{\sqrt{1+\epsilon^2}} = 0.966 \Rightarrow \frac{1}{\epsilon} = 2.736$$

$$y_4 = \frac{1}{2} \left[\left(\sqrt{1 + \frac{1}{\epsilon^2}} + \frac{1}{\epsilon} \right)^{1/4} - \left(\sqrt{1 + \frac{1}{\epsilon^2}} + \frac{1}{\epsilon} \right)^{-1/4} \right]$$

$$= 0.52896$$

$$y_4^2 = 0.27973$$

$$b_1 = 2 y_4 \sin \frac{\pi}{8} = 0.40479$$

$$c_1 = y_4^2 + \cos^2 \frac{\pi}{8} = 1.1332$$

$$b_2 = 2y_4 \sin \frac{3\pi}{8} = 0.9773$$

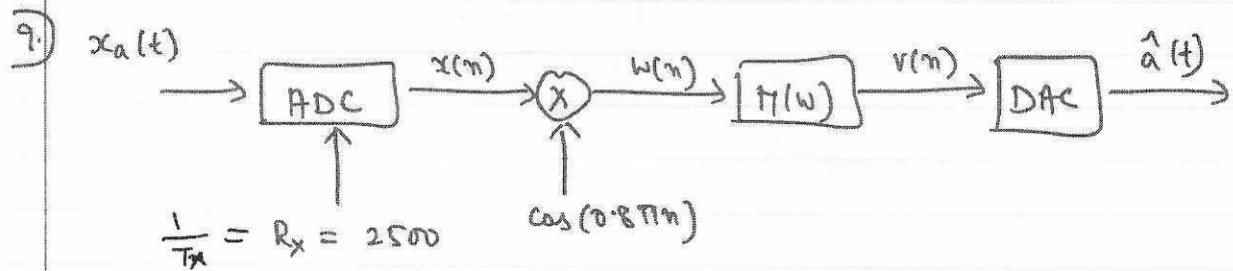
$$C_2 = y_4^2 + \cos^2 \frac{3\pi}{8} = 0.4261$$

$$H_A(s) = \frac{1.133 \times 0.426 \times 0.966}{(s^2 + 0.405s + 1.133)(s^2 + 0.977s + 0.426)}$$

Substitute $s \rightarrow \frac{B\Delta}{\Delta^2 + A_0^2} \Rightarrow \frac{759.456\Delta}{\Delta^2 + (213.765)^2}$

then

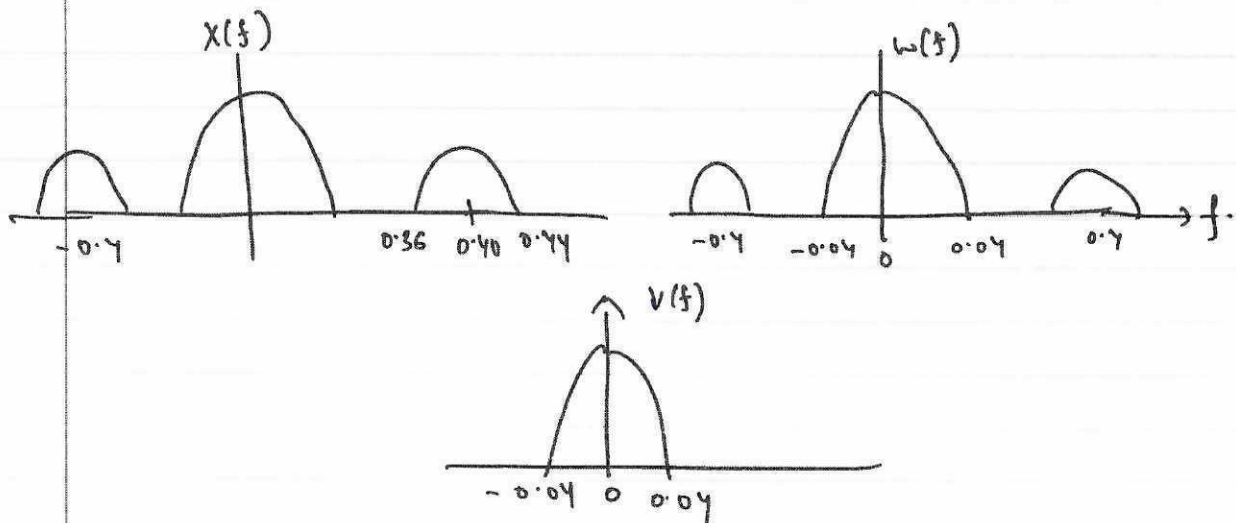
$$\Delta = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} \rightarrow \underline{\underline{H(z)}}$$



$x_a(t) = a(t)$ has 2000π rad/s is bandlimited to $900 \leq F \leq 1100$ Hz.

- Determine & sketch spectra for $x(n)$ & $w(n)$
- Use a Hamming window of length $M=31$ to design a LP linear phase FIR filter $H(w)$ that passes $\{a(n)\}$
- Determine sampling rate of ADC that would allow us to eliminate the freq. conversion

Soln (c) $f_L = \frac{900}{2500} = 0.36$; $f_H = \frac{1100}{2500} = 0.44$



(b) Ideal LPF has passband $-0.04 \leq f \leq 0.04$

$$H_d(\omega) = \begin{cases} 1 \cdot e^{-j15\omega} & ; -0.08\pi \leq \omega \leq 0.08\pi \\ 0, & \text{otherwise} \end{cases}$$

$$h_d(n) = \frac{1}{2\pi} \int_{-0.08\pi}^{0.08\pi} e^{-j15\omega} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-0.08\pi}^{0.08\pi} e^{j\omega(n-15)} d\omega = \frac{2 \sin 0.08\pi(n-15)}{(n-15)}$$

$$h(n) = h_d(n) w_H(n) = 0.54 - 0.46 \cos \frac{2\pi(n-15)}{30}$$

(c) $R_x = 1000$.

8.8.2 Gray-Markel Method

The method of realizing an IIR transfer function $H(z) = P_M(z)/D_M(z)$ consists of two steps. In the first step, an intermediate allpass transfer function $A_M(z) = z^{-M} D_M(z^{-1})/D_M(z) = \tilde{D}_M(z)/D_M(z)$ is realized in the form of a cascaded lattice structure.⁶ A set of independent variables of this structure are then summed in the second step, with appropriate weights to yield the desired numerator $P_M(z)$.

To illustrate the method of realizing the numerator, we consider for simplicity the implementation of a third-order IIR transfer function

$$H(z) = \frac{P_3(z)}{D_3(z)} = \frac{p_0 + p_1 z^{-1} + p_2 z^{-2} + p_3 z^{-3}}{1 + d_1 z^{-1} + d_2 z^{-2} + d_3 z^{-3}}. \quad (8.71)$$

In the first step, we form an intermediate allpass transfer function $A_3(z) = Y_1(z)/X_1(z) = \tilde{D}_3(z)/D_3(z)$. Realization of $A_3(z)$ has been illustrated in Example 8.10, resulting in the structure of Figure 8.32(c). Our objective is to sum the linearly independent signal variables Y_1 , S_1 , S_2 , and S_3 with weights $\{\alpha_i\}$, as shown in Figure 8.39, to arrive at the desired numerator $P_3(z)$. To this end, we need to analyze the digital filter structure of Figure 8.32(c) and determine the transfer functions $S_1(z)/X_1(z)$, $S_2(z)/X_1(z)$, and $S_3(z)/X_1(z)$.

From Eq. (8.69), we have

$$\frac{S_1(z)}{X_1(z)} = \frac{1}{D_3(z)}. \quad (8.72)$$

⁶ $\tilde{D}_M(z)$ denotes the mirror-image polynomial of $D_M(z)$ of degree M ; that is, $\tilde{D}_M(z) = z^{-M} D_M(z^{-1})$.

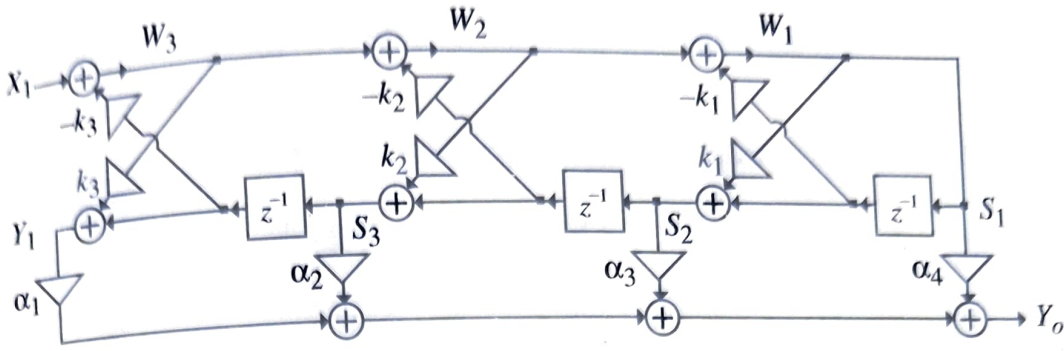


Figure 8.39: The Gray-Markel structure for a third-order transfer function.

Next, we observe from Figure 6.38 that $S_2(z) = (d_1'' + z^{-1})S_1(z)$ and, hence, from the above, we get

$$\frac{S_2(z)}{X_1(z)} = \frac{d_1'' + z^{-1}}{D_3(z)}. \quad (8.73)$$

Finally, from Figure 8.39, we have $S_3(z) = d_2' W_2(z) + z^{-1} S_2(z)$ and $S_1(z) = W_2(z) - d_1'' z^{-1} S_1(z)$. From these equations, the relation $S_2(z) = (d_1'' + z^{-1})S_1(z)$, and Eqs. (8.58a), (8.58b), and (8.60), we arrive at $S_3(z) = (d_2' + d_1' z^{-1} + z^{-2})S_1(z)$, yielding

$$\frac{S_3(z)}{X_1(z)} = \frac{d_2' + d_1' z^{-1} + z^{-2}}{D_3(z)}. \quad (8.74)$$

It should be noted that the numerator of the transfer function $S_i(z)/X_1(z)$ is precisely the numerator of the allpass transfer function $S_i(z)/W_i(z)$.

We now form

$$\frac{Y_o(z)}{X_1(z)} = \alpha_1 \frac{Y_1(z)}{X_1(z)} + \alpha_2 \frac{S_3(z)}{X_1(z)} + \alpha_3 \frac{S_2(z)}{X_1(z)} + \alpha_4 \frac{S_1(z)}{X_1(z)}. \quad (8.75)$$

Substituting Eqs. (8.72) to (8.74) and the expression for $Y_1(z)/X_1(z) = A_3(z)$ as given by Eq. (8.56) in Eq. (8.75), we arrive at

$$\frac{Y_o(z)}{X_1(z)} = \frac{\alpha_1(d_3 + d_2 z^{-1} + d_1 z^{-2} + z^{-3}) + \alpha_2(d_2' + d_1' z^{-1} + z^{-2}) + \alpha_3(d_1'' + z^{-1}) + \alpha_4}{D_3(z)}. \quad (8.76)$$

Comparing the numerators of the right-hand sides of Eqs. (8.71) and (8.76), and by equating the coefficients of like powers of z^{-1} , we thus obtain

$$\begin{aligned} \alpha_1 d_3 + \alpha_2 d_2' + \alpha_3 d_1'' + \alpha_4 &= p_0, \\ \alpha_1 d_2 + \alpha_2 d_1' + \alpha_3 &= p_1, \\ \alpha_1 d_1 + \alpha_2 &= p_2, \\ \alpha_1 &= p_3. \end{aligned}$$

Solving the above, we get

$$\begin{aligned} \alpha_1 &= p_3, \\ \alpha_2 &= p_2 - \alpha_1 d_1, \\ \alpha_3 &= p_1 - \alpha_1 d_2 - \alpha_2 d_1', \\ \alpha_4 &= p_0 - \alpha_1 d_3 - \alpha_2 d_2' - \alpha_3 d_1'', \end{aligned} \quad (8.77)$$